

Harsh Kumar

harshondesk@gmail.com |  harsh-space |  harsh-space |  Harsh00137

Education

- **Bachelor of Technology in Computer Science & Engineering** CGPA: 8.4/10
GL Bajaj Institute of Technology & Management 2027
- **Senior Secondary** Marks: 89.2%
Radiant Academy (CBSE) 2023

Technical Skills

Programming Languages: Python, C++

Data Analysis: NumPy, Pandas, Matplotlib

ML & Deep Learning: Scikit-learn, XGBoost, PyTorch

ML Techniques: Classification, Regression, Anomaly Detection, Feature Engineering, Model Evaluation

Tools & Frameworks: Git, GitHub, VS Code, FastAPI, Firebase

Personal Competencies

Problem-solving · Analytical Thinking · Project Coordination · Team Leadership · Communication

Projects

Jal Sarthi – IoT-based smart water quality monitoring system

ESP32 • Firebase • Flutter

- Led a **3-member team** to build a multi-parameter water-quality monitoring system measuring turbidity, TDS, and temperature; applied **trimmed-mean filtering, multi-point calibration, and temperature compensation**, reducing short-term measurement variance by **30%**
- Implemented **30-second local aggregation and ESP32 buffering** to retain readings during network disruptions, synchronizing averaged data to Firebase Firestore after connectivity was restored

Skillio AI – AI career-learning and skill-gap analysis engine

Python • Scikit-learn • FastAPI

- Built an NLP pipeline using **TF-IDF and cosine similarity** to classify natural-language career goals and identify skill gaps across **6 engineering roles**, achieving **76.7% goal-classification accuracy**
- Built a **DAG** with topological sorting to generate prerequisite-ordered learning paths, distinguishing basic and advanced competencies from detected skill gaps

Coverent – ML-driven parametric insurance platform

Python • XGBoost • Scikit-learn • FastAPI

- Built an **XGBoost risk-scoring model** on **1,500+ synthetic samples** using environmental, behavioral, and historical features to generate **0–100 zone-level risk scores** and drive dynamic weekly premium adjustments
- Implemented **Isolation Forest anomaly detection** using GPS, activity, velocity, duplicate-claim, and historical-pattern signals for automated claim validation

Achievements

- Secured **Rank 248** among **5,000** participants in the HCL AMPlified AI Challenge 2026
- Finalist, myOnsite Healthcare Hackathon 2026
- Semi-finalist, Tata Imagination Challenge 2025 – ranked **top 3%** nationwide

Certifications

- Agentic AI Foundations, Oracle 2026
- Introduction to Machine Learning, Kaggle 2026
- Python Essentials, Cisco Networking Academy 2026

Patent

- **Smart Water Quality Monitoring System using IoT** 2025
Published (India) | Application No. 202511098985 A | Inventor (filed under college)